

ARIA-POC

Unified Perception & Session-Aware Interaction

Architecture Analysis & Implementation Roadmap for a Security/Receptionist Robot Prototype

Prepared by **Fedi Souihi** · TekUp University — Security Robot Project
POC target: personal laptop (verified) → Deployment target: NVIDIA Jetson Orin Nano 8GB

Document	Security Robot — Architecture Analysis & Roadmap (pre-code deliverable)
Version / Status	1.2 — v1.1 revisions + scope decision (see below)
Scope decision	English-only interaction for the POC. Multi-language switching (Arabic / French / Derja) is explicitly out of scope per project decision 2026-09-19 — too unreliable to implement well. Contracts keep a language field so a language can be added later by configuration; Derja is permanently out of scope for this architecture class.
Date	2026-09-19
Primary input	TestExecutionReport_updated2.csv (manual test campaign, MSBOT-141...345)
Design rule	Every capability is a swappable service behind configuration — features are added and removed without rewrites.

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1. Executive summary

The current ARIA robot performs well in its happy path — a single known visitor approaches, is recognized in under 2 seconds, and holds a short cloud-LLM conversation. The manual test campaign, however, shows the design is fragile everywhere else: single-human and group detection are **two separate code paths** (evaluated once per session, not continuously), identity is never re-verified (so a fast visitor swap inherits the previous person's context), departure is a fixed 20-second face-visibility timer, end-of-utterance is a hard-coded pause timeout, cross-session memory does not recall past topics, a camera crash requires restarting the whole application, and the cloud backend is trivially prompt-injectable.

This roadmap replaces that architecture with a **single dynamic perception pipeline** in which zero, one, or many people are handled by the same code path — "group mode" stops existing as a mode and becomes emergent scene state — and wraps it in a service-oriented runtime where **sessions, identity, and memory are robust first-class concepts**. Every capability (detector, tracker, recognizer, STT, TTS, LLM, memory) lives behind an interface registered in a plugin registry, so improvements are config changes, not rewrites, and features can be removed just as easily as they are added.

Proof-of-concept target (verified on this machine): ASUS Vivobook 16 — Intel Core 5 210H (8C/12T), NVIDIA RTX 4050 Laptop (6 GB VRAM), 16 GB RAM, Windows 11. **Deployment target:** Jetson Orin Nano 8 GB under JetPack 6.x. All model and memory budgets are chosen to fit both platforms; the Resource Governor enforces them at runtime.

Deliverable order: documents first (this report, plus the repository-level ARCHITECTURE.md), then implementation phases 0–7, each ending in a demo-able, testable state. The phase ordering is feature-gated, not date-gated.

Success criteria

- **One code path:** 0/1/N people produce identical behavior contracts; no mode flags exist anywhere in the code.
- **Continuous identity:** a session is bound to a verified person-track and re-checked periodically; a fast A→B handoff can never transfer A's context to B (fixes NFR-07 Step 4, NFR-01 Step 3).
- **Working recall:** a returning visitor is greeted with their last open topic — "Did you find the library last time?" (fixes FUNC-10 Steps 1–2).
- **Natural audio:** adaptive end-of-utterance replaces the hard-coded pause timeout; non-directed noise and overheard speech are gated out (fixes FUNC-14 Steps 8, 11, 12).
- **Resource discipline:** everything fits the laptop budget; memory returns to idle baseline after sessions (fixes NFR-05 residue).
- **Self-healing:** no camera/mic failure requires restarting the application (fixes NFR-03 Step 6); camera tampering is detected (fixes FUNC-05 Step 13).
- **Hardened conversation:** system prompt extraction is mitigated by a no-secrets design plus input/output screening (fixes NFR-06).
- **Built on telemetry (new requirement):** structured logging and the event timeline exist from Phase 0 — any malfunction must be reproducible from the robot's own logs (§9).
- **Built on tests (new requirement):** contract tests, hermetic fakes, and deterministic fixture replays from the first commit, so features that misbehave later are reproduced and isolated in minutes (§10).

2. Inputs — findings from the test report

The manual campaign (`TestExecutionReport_updated2.csv`) is the empirical foundation of this prototype. Each finding below is treated as a requirement, not a bug ticket.

Ref	Area	Finding (as tested)	Prototype response
FUNC-14 S8, S10	End of utterance	Pause detection is a hard-coded timeout; natural mid-sentence pauses trigger premature responses.	VAD segmentation + semantic end-of-turn detection (Smart Turn, Arabic/French included) with threshold fallback (§5.1, Phase 2).
FUNC-14 S11–12	Noise / directionality	Loud non-speech noise can be transcribed; any heard speech is treated as directed at the robot.	VAD + engagement gating + voice-print check against the engaged identity (Phase 2).
FUNC-10 S1–2	Cross-session memory	Same-day and next-day topic recall not functional.	MemoryCard extraction on session close + recall composer on re-engagement (Phase 4).
FUNC-07 S6–7, S13	Multilingual / diarization	Language detection resets per utterance and misfires on short input; no mid-sentence AR→FR code-switching; overlapping speakers merged into one transcription.	Out of scope (project decision): English-only POC; no language detection/switching. The language field stays in contracts for a later config-only addition. Overlap (S13) keeps the one-at-a-time prompt (Phase 2).
FUNC-06 S2, S8–9	Face recognition	False-positive match of an unregistered tester; printed-photo and on-screen spoofs both accepted.	Multi-frame identity voting with margin; passive anti-spoofing as a Phase 7 research item.
FUNC-05 S3–4, S8, S13	Face detection	Full profile and downward gaze fail (gaze-down reads as "user left"); masks defeat detection; no camera-tamper awareness.	Track-based presence (person body, not face-only) + frame-statistics tamper detector (Phase 1).
NFR-01 S3	Identity handoff	No ID-locking: a second person entering during an active session inherits the first person's identity; timer resets only on face visibility.	Session bound to a track identity, re-verified on an interval; departure is multimodal (Phase 1/3).
NFR-03 S2, S6	Peripheral failure	Camera module crash needs a full restart of <code>main.py</code> ; reconnecting hardware is not detected.	Per-service watchdog + device monitor with hot reconnect (Phase 5).
NFR-05 S1–3	Resource hygiene	~800 MB idle, 910 MB during interaction; +300–400 MB never reclaimed after sessions.	Bounded queues, session teardown, Resource Governor, soak-test gate (Phase 0/5).
NFR-06 S2–5	Prompt injection	Direct injection, DAN roleplay, and translation-framed extraction all succeed against the cloud backend.	No-secrets prompt design, instruction hierarchy, input/output screening (Phase 3).
NFR-07 S1, S4	Data exposure	Sensitive-ID masking has trigger edge cases; context theft via session swap succeeds.	Masking at the policy layer with regression tests; identity re-verification closes the swap hole (Phase 1/3/4).
FUNC-11 (all)	Group interaction	Group behavior untested — by design the system has no group awareness.	SceneManager + AttentionPolicy make multi-person handling a first-class state (Phase 1/3).
FUNC-04 (all)	LiDAR presence	Blocked / untested.	Camera-based presence for the POC; LiDAR re-enters later as a pluggable sensor service (Phase 7).

3. Target hardware

3.1 POC platform — verified on this machine

Component	Detail	Design implication
Machine	ASUS Vivobook 16 V3607VU	Consumer laptop; thermal headroom adequate for short inference bursts.
CPU	Intel Core 5 210H — 8 cores / 12 threads @ 2.2 GHz base	Enough threads to offload STT or TTS to CPU when VRAM is tight.
GPU	NVIDIA GeForce RTX 4050 Laptop — 6 GB VRAM	Binding constraint. Every model budget sized against ~6 GB with OS headroom.
RAM	16 GB	Comfortable for the full service stack plus OS.
OS	Windows 11 Home, 64-bit	Native Windows stack (no WSL requirement); Edge available for tooling.

3.2 Deployment platform — Jetson Orin Nano 8 GB

Component	Detail
Compute	1024-core Ampere GPU (32 Tensor cores) + 6× Cortex-A78AE; up to 40 TOPS (sparse INT8).
Memory	8 GB LPDDR5, unified — CPU and GPU share one pool; every "VRAM" byte is taken from the same 8 GB the OS needs.
Software	JetPack 6.x (CUDA, TensorRT, ONNX-Runtime path); models exported to TensorRT INT8 in Phase 7.

Implication. The laptop's split budgets (6 GB VRAM + 16 GB RAM) are strictly more forgiving than the Orin's single 8 GB pool, so anything that fits the budget table in §4 will also fit the Orin with the same architecture — only the model format (TensorRT engines) and TTS/LLM defaults change per profile.

4. Resource budget & governor policy

4.1 Model budget (POC laptop)

Component	Selection	Budget	Notes
Person detection	YOLO26n @ 640 (person class only)	~0.5 GB VRAM	Jan-2026 edge-first family: NMS-free end-to-end head, +0.6–0.8 mAP over the YOLO11 class; pose variant available for future facing-estimation.
Face detection	SCRFD-500M (alt: OpenCV YuNet)	~0.1 GB	Native multi-face in one pass.
Face embedding	ArcFace r50 (InsightFace buffalo_1)	~0.3 GB	512-d vectors; runs per-track, not per-frame.
VAD + turn detection	Silero VAD v5 + Smart Turn v3.2 (ONNX)	~0.1 GB	Silero (~2 MB) streams continuously; Smart Turn (8 MB int8, CPU) runs only during silence to decide end-of-turn.
STT	faster-whisper_large-v3-turbo int8_float16 (CTranslate2)	~1.2 GB (or CPU)	Near-large-v3 accuracy at turbo cost; small stays as the CPU / tight-VRAM fallback (12 threads available).
LLM (local fallback)	Qwen3-4B-Instruct-2507 Q4 via Ollama	~2.5 GB on demand	GPT-4.1-nano-class benchmarks, strong multilingual; loaded only when cloud is unavailable/timeout.
LLM (default)	Cloud completion API	0 VRAM	Parity with current robot's backend class.
TTS	Piper1-GPL (CPU) · Kokoro-82M (EN/FR) · edge-tts (cloud)	0 VRAM	CPU-only or network; never competes for VRAM; Kokoro adds near-studio EN/FR offline quality.
Total steady-state (cloud LLM)		~2.2 GB	Headroom for session telemetry, overlay, and spikes.
Peak (local LLM active)		~4.4 GB	Inside 6 GB with OS headroom; STT demotes to CPU under VRAM pressure (§4.2).

4.2 Governor policies (enforced at runtime from Phase 0)

- **Camera pacing:** 15 fps while the scene is active (tracks present / session open), 2 fps idle; resolution 1280→640 downscale under pressure. Detection cost tracks scene activity, not wall-clock.
- **Placement:** STT runs on GPU by default and is demoted to CPU when VRAM pressure crosses a threshold; local LLM loads only on demand and unloads after a cooldown.
- **Backpressure:** all queues are bounded; under overload the governor drops *frames*, never audio, and never blocks the audio thread behind vision work.
- **Teardown:** on session close, per-session objects, caches, and queue buffers are released; a soak test (Phase 5) asserts RSS returns to within ~10% of the idle baseline — the direct answer to NFR-05's +300–400 MB residue.

5. Technology selections & rationale

Every selection sits behind a domain interface, so an alternative is a configuration change rather than a rewrite. "Research spike" items are deliberately scheduled, not hand-waved.

Domain	Selection	Why	Alternatives (swap path)
Detection + tracking	Ultralytics YOLO26n + ByteTrack	Current-generation (Jan 2026) edge-first family: native NMS-free end-to-end inference, lighter head (no DFL), ~43% faster CPU inference than the YOLO11 class at equal-or-better mAP; a pose variant exists for future facing/engagement estimation. ByteTrack stays the fastest solid baseline and tracker choice is a YAML file, not code.	BoT-SORT (camera-motion comp, optional ReID), OC-SORT, Deep OC-SORT, TrackTrack (new Ultralytics default) — all drop-in config alternates. Phase 1 spike benchmarks on recorded fixtures. YOLO11n remains a fallback weight.
Face pipeline	InsightFace buffalo_1 : SCRFD detector + ArcFace r50 (512-d)	Multi-face detection in one pass; strong embeddings; per-track gallery + multi-frame voting closes the false-positive class seen in testing.	OpenCV YuNet + ONNX ArcFace (smaller footprint); FaceNet (legacy).
VAD / EOU	Silero VAD v5 + Smart Turn v3.2	Silero (~2 MB) segments speech robustly; Smart Turn (pipecat-ai, BSD-2, Whisper-Tiny backbone, 8 MB int8) then decides <i>semantically</i> whether the turn is complete — 23 languages including Arabic and French, ~10–100 ms on CPU, audio-native (uses prosody, not just text).	WebRTC VAD (lighter, less robust); plain threshold mode as degenerate config fallback; LLM-assisted endpointing as a later upgrade.
STT	faster-whisper large-v3-turbo int8_float16	Large-v3-class accuracy at turbo cost (~8× less compute than large-v3); officially supported in faster-whisper with batched inference and a built-in Silero VAD filter. English-first per scope decision — the model stays multilingual, so a language could be added later by config.	small int8 as the CPU / tight-VRAM fallback; whisper.cpp (Vulkan); UFAL whisper_streaming as an optional continuous-streaming policy layer.
Voice-print	SpeechBrain ECAPA-TDNN x-vectors	Cheap speaker embedding used to gate "is this speech from the engaged person?" and reject overheard speech; groundwork for diarization.	pyannote embeddings (heavier); simple energy heuristics as degenerate mode.
TTS	TTSBackend : edge-tts (cloud, default) · Kokoro-82M (offline EN) · Piper1-GPL (offline, multilingual capability)	English-first per scope decision: edge-tts for the laptop demo; Kokoro-82M (Apache, 82M params) as the strongest offline English voice. Piper1-GPL ships French/Arabic voices (fr_FR , ar_J0-kareem) — capability retained behind the same interface, not wired into the POC.	XTTS rejected (license/weight). Adding a backend or voice = one class + config entry.
LLM	ChatProvider : cloud API (default) + Ollama Qwen3-4B-Instruct-2507 Q4 (fallback)	Cloud keeps parity with the current robot's quality; the local fallback is a 4B Apache model that matches GPT-4.1-nano on public benchmarks with strong multilingual scores (incl. Arabic) and 256K context — a class above the Qwen2.5-3B originally shortlisted.	Qwen3-1.7B/0.6B for guardrail classification or tighter budgets; larger cloud tiers by config.
Memory	SQLite (WAL) + optional sqlite-vec	Zero-service dependency, file-portable, Jetson-friendly; vector index is an optional layer, not a service.	ChromaDB/FAISS only if semantic recall demands it later.

Runtime	Python 3.11, asyncio, OpenCV, onnxruntime-gpu, psutil+NVML	Single-language stack with CUDA access on Windows and Jetson alike.	TensorRT/DeepStream integration is a Phase 7 optimization, not a re-architecture.
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5.1 End-of-turn detection, three tiers (replaces the hard-coded timeout)

- **Tier 1 — VAD (always on):** Silero VAD v5 streams at 16 kHz and segments speech; non-speech events (doors, coughs, footsteps) never open a turn in the first place (FUNC-14 Step 11).
- **Tier 2 — semantic turn decision (default):** when VAD reports silence, the buffered turn audio (≤ 8 s, trailing-aligned) is scored by **Smart Turn v3.2** — an open-source, 8 MB int8 CPU model (Whisper-Tiny backbone) purpose-built for "should the agent respond yet?". It distinguishes "I'm looking for... the library" (pause, keep listening) from "...the library." (complete) — the exact FUNC-14 Step 8 failure case — using prosody, not just timing, in 23 languages including Arabic and French.
- **Tier 3 — fallback:** if Smart Turn is disabled or unavailable, a configurable silence threshold (default 600 ms) takes over, so behavior never drops below today's baseline. A max-utterance cap (default 20 s) prevents runaway capture (preserves the Step 10 behavior the tests liked).

5.2 Language scope — English-only by decision

Per the project decision of 2026-09-19, multi-language interaction — Arabic, French, and especially Derja — is **out of scope** for the POC: too unreliable to implement well on this hardware class, and a half-working version is worse than none. The robot interacts in English only; there is no language detection, no session-language prior, no switching logic. Nothing about the architecture forecloses the alternative: Whisper and the LLM remain language-agnostic, and every data contract (`Utterance` , `Session`) carries a language field — adding French or Arabic later is a configuration matter (STT model, TTS voices, prompt), not a redesign. Derja specifically is documented as permanently out of scope for this architecture class.

5.3 Identity voting (fixes the false-positive match)

A face match is only *provisional* until the same embedding wins with margin over the runner-up across $\geq N$ frames (default 3) within a track. The track accumulates an embedding gallery with an EMA centroid, so brief pose/lighting variation does not poison identity. Enrollment is a small tool that captures a gallery per person.

5.4 LLM fallback router

Cloud calls are wrapped with a timeout; on timeout or network loss the router loads the local **Qwen3-4B-Instruct-2507** (~2.5 GB Q4; first call pays a load latency, later calls are fast; unloads after a cooldown), answers with a localized "offline mode" notice if needed, and retries cloud afterwards. Guardrail classification can share the runtime with a Qwen3-0.6B/1.7B tier if heuristic screening proves insufficient. This upgrades NFR-03 Step 4 from "technical difficulties" to graceful degradation.

6. System architecture



6.1 Core primitives (built once in Phase 0, used by everything)

Primitive	Responsibility
EventBus	Typed publish/subscribe (TrackAppeared , TrackLost , FaceMatched , UtteranceHeard , TamperDetected , DeviceLost / DeviceRestored , SessionOpened/Closed , ...) with bounded queues and an explicit backpressure policy. Consumers and publishers never import each other.
Service	Base class: lifecycle (init/start/stop/health), heartbeat, declared config schema, declared events produced/consumed. Every feature is one of these.
PluginRegistry	Discovers and instantiates services from YAML config; enabling/disabling/replacing a capability never touches code outside that capability.
Watchdog	Supervises heartbeats; restarts a crashed service (e.g., the camera module) without restarting the application.
ResourceGovernor	Watches VRAM/RAM/CPU; applies pacing/placement policy (§4.2); owns teardown semantics.
DeviceMonitor	Watches camera/mic enumeration; publishes device events; services react with hot reconnect.

6.2 Data contracts

Contract	Key fields
Track	id, bbox, velocity, engagement state, age, last-seen.
PersonIdentity	embedding centroid + gallery, profile ref, name (if enrolled), confidence, verification timestamp.
Utterance	text, language, confidence, voice-print ref, start/end timestamps, gating verdict.
Session	bound identity, FSM state, history ref, opened/closed timestamps, language.
MemoryCard	topic, summary, status (open/resolved), created/last-touched timestamps, profile ref.

7. Unified detection design

This is the direct answer to the project's core problem: two fragile code paths (single vs group) with one-shot evaluation. The prototype has **one** pipeline; mode distinctions do not exist.

#	Design decision	Mechanism
1	Sense once, decide continuously	Every frame → person detection → tracker → N persistent Tracks. The scene is re-evaluated every cycle, not once per session. A "session" is simply a relationship between the robot and engaged tracks.
2	Per-track state machine	PRESENCE → NEAR → ENGAGED → DEPARTED , driven by proximity, dwell time, facing/pose, and speech. Replaces the single-person trigger assumption; camera-based presence covers the LiDAR role for the POC.
3	Identity binding with ID-lock	Face pipeline runs per track (not per frame): detect → embed → gallery vote → PersonIdentity . Once bound, the session is locked to that identity.
4	Continuous re-verification	The bound face is re-verified every N seconds (default 5, configurable) and on suspicion triggers (bbox jump, a new face overlapping the engaged track). If the bound face is gone while a different face occupies the track → session freezes and re-authenticates; context never transfers (closes the NFR-07 Step 4 theft path).
5	Multimodal departure	Departure requires face-loss and no voice activity and no recent interaction; any activity resets the absence timer. Replaces the fixed 20 s face-only timer (NFR-01 Step 3) and the "look down = left" failure (FUNC-05 Step 4).
6	Group awareness by policy	SceneManager publishes present/near/engaged people; the Dialogue Manager's AttentionPolicy selects the addressee (configurable: nearest-facing, most-recent-speaker, first-arriver), acknowledges bystanders, and treats A→B handoff as a first-class transition (FUNC-11).
7	Tamper & failure awareness	Frame-statistics detector (uniformity/blur/blackout) publishes TamperDetected (FUNC-05 Step 13); DeviceMonitor events drive hot reconnect (NFR-03 Step 6).

Acceptance invariant: for any $N = 0, 1, 2, \dots$ people in frame, the perception layer emits the same event vocabulary and the same per-track state machine applies. Tests assert behavioral parity rather than separate single/group test suites — the defect class "two code paths diverge" becomes structurally impossible.

8. Sessions & personalized memory

8.1 Lifecycle

- **Open:** a track reaches `ENGAGED` and identity is bound → `SessionOpened`. Greeting is composed from the profile (name if enrolled; recall content if any).
- **During:** raw turns kept in a bounded FIFO (default 6, configurable) *plus* a running LLM-compressed summary — context survives beyond the FIFO limit, an explicit improvement over the current 5-exchange ceiling.
- **Close:** departure confirmed (multimodal, §7) or farewell → the LLM extracts **MemoryCards** from the conversation: topic, short summary, open/resolved status. Cards are appended to the profile.
- **Re-engage:** the RecallComposer injects the most recent and any *unresolved* cards into the greeting context → "Did you find the library last time?" (fixes FUNC-10 Steps 1–2 for same-day and next-day gaps alike, since cards are timestamped).

8.2 Profile model

Field group	Contents
Identity	Face embedding centroid + recent gallery, enrolled name (if any), temporary flag for strangers.
History	Session list with timestamps, MemoryCards, topics discussed.
Preferences	Language, voice preference, opt-out flags.
Biometrics-adjacent	Voice-print reference (for gating only), last-seen.

8.3 Privacy & retention

- Sensitive-ID masking is applied at the **policy layer** — a regex + context filter over both TTS text and display output — with regression tests covering the trigger edge cases found in NFR-07 Step 1.
- Session data is wiped on *confirmed departure*, not only when a replacement session starts — closing the lingering-context gap in NFR-07/FUNC-10.
- Strangers' temporary profiles auto-expire (default 30 days); any profile can be deleted on request; raw audio is never persisted beyond the active utterance.

9. Observability & logging from day one

Logging is a Phase-0 deliverable, not a retro-fit. The governing rule: **any malfunction observed at the robot must be reproducible from its own telemetry** — logs plus the event timeline — without re-running the scenario. Four layers:

Layer	What it captures	How it is used
Structured logs	One logger per service; every line is JSON carrying <code>ts</code> , <code>level</code> , <code>service</code> , <code>event</code> , <code>session_id</code> , <code>track_id</code> , <code>utterance_id</code> bound automatically as context. Pretty human-readable console mode for development.	Failures in <code>aria.audio.stt</code> are greppable without reading vision noise; log levels are set per module from config; every state transition that matters logs one line explaining <i>why</i> .
Event timeline ("flight recorder")	The EventBus taps every event into a JSONL timeline; an in-memory ring buffer of the last ~5 minutes is auto-dumped to disk on anomalies (service crash, watchdog restart, governor throttling, tamper detection).	Post-incident reconstruction: load the timeline with <code>tools/timeline_view.py</code> and see exactly what perception saw, what the dialogue decided, and when — no re-staging of people or hardware.
Metrics	Counters and histograms: frame rate, detection latency, track churn, end-of-turn decision latency, STT real-time factor, TTS synthesis latency, LLM time-to-first-token, VRAM/RAM.	Drives the Phase 6 dashboard and the budget assertions (§4); every governor decision is logged together with the metric value that triggered it.
Privacy in telemetry	The same masking policy layer (§8.3) applies to logs and the timeline; IDs and names are redacted by the shared filter; raw audio/video never touches disk except in explicitly recorded fixtures.	Debuggability without creating a new data-leak channel — consistent with the NFR-07 posture.

Phase-0 rule of thumb baked into code: every governance decision and state transition (session open/close, identity re-verification, language switch, device loss, fallback activation) emits exactly one log line that, read alone, explains the decision — inputs, thresholds, chosen value.

10. Test-first engineering

The same event-bus boundaries that keep features swappable also make them individually testable. The rule enforced from the first commit: **no feature is merged without a test that fails without it**. Testing is how the robot stays debuggable months later — when something malfunctions on site, the workflow below reproduces it deterministically.

Test type	Mechanism	What it protects
Contract tests (per service)	Each interface (Detector , Tracker , FaceRecognizer , STT , TTSTBackend , ChatProvider , MemoryStore) has a contract suite; real and fake implementations must pass the same suite.	Swapping a backend cannot silently change behavior; a malfunctioning feature is isolated to one implementation in one step.
Fixture replays	tools/record_fixture.py records real camera+mic sessions once (single visitor, group, A→B handoff, occlusion, gaze-down, noisy room); tools/replay.py feeds them deterministically through the pipeline.	Full-stack scenarios are reproducible without hardware, people, or cloud — the substrate for debugging future regressions.
Scenario tests (CSV-derived)	Each mapped defect row (§14) becomes an expected event sequence over a fixture — e.g., "mid-sentence pause → no response attempt until Smart Turn says complete".	The original manual test campaign becomes an executable acceptance suite; old failures resurface automatically if regressed.
Hermetic fakes	FakeCamera , FakeMic , FakeLLM (scripted replies + canned latencies), FakeTTS ; a "test" config profile wires every source to a fake so the whole stack runs headless.	Dialogue FSM, attention policy, memory, and guardrails are unit-testable in isolation with zero network or GPU.
Golden files	Transcripts, identity-vote outcomes, and MemoryCard extractions are pinned to golden files with explicit tolerance flags where nondeterminism is legitimate.	Model upgrades show a reviewable behavioral diff instead of surprising users.
Benchmarks-as-tests	bench_vision / bench_audio assert the budget thresholds of \$4; a soak test asserts RSS recovery (NFR-05).	Resource discipline is enforced in CI-lite, not hoped for.

Debug workflow for future malfunctions

Reproduce (fixture replay or auto-dumped timeline) → isolate (per-service contract tests narrow it to one implementation) → fix → the scenario joins the permanent regression suite. Because the failure path is replayable, "it works differently now" disagreements end with evidence, not opinion.

11. Guardrails & robustness

11.1 Prompt-injection defense (NFR-06)

The test campaign showed the cloud backend fully extractable via direct, roleplay, and translation-framed attacks. The prototype assumes cloud models cannot be fully trusted and applies defense-in-depth:

- **No secrets in the prompt:** behavioral rules are enforced in code (policy layer), so a successful extraction yields nothing critical.
- **Input screening:** heuristic classifier for injection patterns (instruction-override phrasing, role-reset framing, encoded/translation-framed extraction) flags or rewrites before the LLM sees the text.
- **Instruction hierarchy + canary:** system prompt includes layered instructions and a canary token; the output filter drops responses containing the canary or policy keywords.
- **Zero agency:** no tool, file, or network access from the model — the attack surface stays in-context only, by construction.

11.2 Operational robustness

- **Watchdog self-heal:** a crashed camera service is restarted in place; the application keeps running (direct fix for NFR-03 Step 6's "full restart of main.py").
- **Hot device reconnect:** `DeviceMonitor` detects camera/mic re-enumeration and the affected service rebinds without restart.
- **Degradation modes:** vision-only (mic lost), audio-only (camera lost/tampered), offline-brain (network lost → local LLM). The robot states which mode it is in, on-screen and aloud.
- **Memory hygiene:** bounded queues + teardown + governor (§4.2); a 60-minute mixed-use soak test with RSS-recovery assertion gates Phase 5.
- **Startup correctness:** audio devices are selected by config identity (name match with fallback), removing the post-reboot "change audio in/out" manual step seen in NFR-04 Step 1.

12. Repository layout & modularity rules

```
aria-poc/
ROADMAP.md · ARCHITECTURE.md · README.md
configs/
  laptop.yaml · jetson.yaml · features/*.yaml
aria/
  core/          event_bus · service · registry · config · watchdog · governor · telemetry (logging + timeline + metrics)
  perception/    camera · detector · tracker · face · scene · tamper
  audio/         mic · vad · turn (smart-turn) · stt · lid · speaker_id
  cognition/     dialogue · attention · guardrails · providers/
  expression/    tts/ · display
  memory/        profiles · sessions · cards · recall
  tests/         contracts/ (per-interface suites) · scenarios/ (CSV-derived) · fixtures/ · golden/
  tools/         record_fixture.py · replay.py · bench_vision.py · bench_audio.py · timeline_view.py · dashboard.py
```

Modularity rules (how expansion stays easy)

- **Feature = Service + config entry.** Adding a capability: implement `Service`, declare its config schema and the events it produces/consumes, register in YAML. Removing a capability: disable its config entry (or delete the folder) — nothing else changes.
- **One interface per domain:** `Detector`, `Tracker`, `FaceRecognizer`, `STT`, `TTSBackend`, `ChatProvider`, `MemoryStore`. Improvements arrive as new implementations behind the same interface.
- **Events are the contract:** the event registry is documented; adding a consumer never requires touching publishers, which is what keeps perception, cognition, and expression independently evolvable.
- **Profiles over forks:** laptop and Jetson differ by config profile (models, placement, defaults), never by code branches.
- **Test tooling is product code:** fixture recording/replay exists from Phase 1, so every later phase is validated against identical deterministic scenes.

13. Phases 0–7 with exit criteria

Phase	Scope	Exit criteria
0 — Skeleton	Repo scaffold; ROADMAP.md + ARCHITECTURE.md; core primitives (EventBus, Service, Registry, Config, Watchdog, Governor); structured logging + event timeline recorder (\$9) ; test scaffolding (\$10) : contract-test harness, hermetic fakes, fixture recorder.	Services start/stop/restart cleanly from config; heartbeat supervision demonstrable; bounded-queue backpressure tested; every service emits structured logs with correlation ids; the contract suite passes green on fakes.
1 — Unified Perception	Camera service (hot-reconnect) + YOLO26n + ByteTrack → SceneManager; FacePipeline (SCRFD + ArcFace, voting, gallery); tamper detector; debug overlay with track/identity HUD; first fixture recordings (single, group, handoff, occlusion).	0/1/N people via one code path; track survives 3 s occlusion; identity stable across re-approach; periodic re-verification demonstrable; bench_vision within \$4 budget; scenario tests run against fixtures, not live people.
2 — Audio	Mic + device monitor; Silero VAD + Smart Turn end-of-turn (three-tier, \$5.1); faster-whisper large-v3-turbo (English-first per scope decision); ECAPA voice-prints; TTS abstraction (Piper1-GPL + Kokoro + edge-tts); barge-in v1 (duck on speech).	FUNC-14 Steps 8/10/11/12 behaviors fixed in replay tests; measured EOU latency and Smart-Turn false-response rate on pause fixtures; utterance gating verified end-to-end.
3 — Cognition	Dialogue FSM (IDLE→GREETING→CONVERSING→HANDOFF→FAREWELL→CLOSED); AttentionPolicy; ChatProvider (cloud + Ollama fallback router); Guardrails + ID masking.	Full conversation on laptop; NFR-06 probes mitigated (no secrets leak, canary filter works); network cut → local model answers.
4 — Memory	ProfileStore; MemoryCards; RecallComposer; retention/expiry and wipe semantics.	FUNC-10 recall scenario passes (same-day + next-day via timestamps); masking edge-case regression tests pass.
5 — Robustness	Watchdog self-heal; hot device reconnect; degradation modes; Resource Governor enforcement; soak tests.	Camera crash heals without app restart (NFR-03); RSS returns to baseline after sessions (NFR-05); device replug detected live.
6 — Validation	CSV-derived test matrix as scripted + manual checklist; fixture replay harness; latency/memory dashboard; internship demo mode.	All mapped defect rows (\$14) demonstrated fixed or explicitly deferred with reasons; demo runs end-to-end on the laptop.
7 — Jetson track	TensorRT INT8 export notes; JetPack deployment guide; mic-array/directionality research; diarization; passive anti-spoofing; Derja STT research; LiDAR re-integration as sensor service.	Deployment guide validated on-device (when hardware access is granted); research items produce go/no-go decisions.

Phases 0–6 target the laptop and are sequenced so each phase ends with something demo-able. Phase 7 is designed now (interfaces and config reserved) but executed when Jetson access is available.

14. Defect → feature → phase map

CSV ref	Defect / observation	Prototype feature	Phase
FUNC-14 S8	Hard-coded pause timeout cuts users off mid-sentence.	Three-tier turn detection: VAD → Smart Turn → threshold fallback.	2
FUNC-14 S11–12	Non-speech noise transcribed; overheard speech answered.	VAD + engagement gating + voice-print check.	2
FUNC-14 S10	Silence timeout works but is fixed and opaque.	Tiered turn-detection policy, documented and tunable.	2
FUNC-10 S1–2	No same-day/next-day topic recall.	MemoryCards + RecallComposer.	4
FUNC-07 S7	Short input flips language ("hello" → German).	Out of scope — English-only POC (project decision).	—
FUNC-07 S6	Mid-sentence AR→FR code-switching unhandled.	Out of scope (project decision); documented in §5.2.	—
FUNC-07 S13	Overlapping speakers merged into one transcript.	Polite one-at-a-time prompt; diarization stays a research item.	2/7
FUNC-06 S2	Unregistered tester matched to a registered profile.	Multi-frame identity voting with margin.	1
FUNC-06 S8–9	Photo/screen spoof accepted.	Passive liveness research (stretch).	7
FUNC-05 S3–4, S8	Profile/gaze-down/mask detection failures; gaze-down reads as departure.	Body-track presence + multimodal departure.	1
FUNC-05 S13	No camera-tamper awareness.	Frame-statistics tamper detector + alert.	1
FUNC-11	No group interaction capability.	SceneManager + AttentionPolicy; group tests from the CSV become acceptance tests.	1/3/6
FUNC-04	LiDAR untested (blocked).	Camera presence for POC; LiDAR as pluggable sensor later.	7
NFR-01 S3	Fixed 20 s face-only departure; no ID-locking; wrong-person inheritance.	Multimodal departure + ID-lock + re-verification.	1/3
NFR-03 S2/S6	Camera crash needs full restart; replug not detected.	Watchdog self-heal + DeviceMonitor hot reconnect.	5
NFR-03 S4	Network loss → "technical difficulties".	Local LLM fallback router.	3
NFR-04 S1	Manual audio device reselection after reboot.	Config-pinned device selection with fallback.	2/5
NFR-05 S1–3	+300–400 MB never reclaimed after sessions.	Bounded queues, teardown, governor, soak-test gate.	0/5
NFR-06 S2–5	System prompt fully extractable via injection.	No-secrets design + input screening + canary output filter.	3
NFR-07 S1	Sensitive-ID masking trigger edge cases.	Policy-layer masking with regression tests.	3/4
NFR-07 S4 / FUNC-06 note	Fast A→B swap inherits A's context (context theft).	Identity re-verification + freeze-and-re-auth on suspicion.	1/3
FUNC-13 S1	No live interruption by design.	Streaming pipeline + barge-in v1 (duck/pause TTS).	2

15. Assumptions, limitations, risks

15.1 Assumptions

- Internet is available on the laptop for cloud LLM/TTS defaults; the design is fully offline-capable by config (Piper1-GPL + Kokoro + Ollama) for the Jetson target.
- POC uses the laptop webcam + a headset mic; `CameraService` abstracts the source (webcam / CSI / recorded file / replay), so the robot's real sensors plug in without re-architecture.
- The current robot uses a CSI camera + USB mic + LiDAR + screen; all are represented as swappable services in the target architecture.
- Jetson hardware access comes later; Phase 7 is designed now and executed then.

15.2 Known limitations (documented, not hidden)

- Multi-language interaction (Arabic, French, Derja) is deliberately **out of scope** per project decision — English-only POC. Contracts keep language fields so a language can be added later by configuration; Derja is permanently out of scope for this architecture class.
- Overlapping-speaker separation is not in the POC's first milestone; the system politely requests one-at-a-time speech (the human-appropriate version of diarization-lite).
- Passive anti-spoofing (photo/screen liveness) is research, not a POC-phase guarantee.
- Cloud prompt injection can be mitigated, not eliminated — the no-secrets design bounds the blast radius and the residual risk is documented in ARCHITECTURE.md.

15.3 Risks & mitigations

Risk	Impact	Mitigation
6 GB VRAM proves tighter than budgeted with the local LLM loaded.	Fallback path latency or OOM.	Governor demotes STT to CPU; LLM loads on demand only; budget re-benchmarked in Phase 1 before Phase 3 commits to the local model.
Tracker choice underperforms on real scenes (ID swaps).	Identity churn breaks sessions.	Tracker is a YAML swap; Phase 1 spike benchmarks ByteTrack vs BoT-SORT (with ReID) on recorded fixtures before freeze.
Whisper LID margin policy feels wrong in practice.	Language flip-flops or sticks wrongly.	Margin and prior are runtime-configurable; logged per-utterance for tuning with real usage.
Windows-specific audio/camera quirks slow the POC.	Schedule slip.	Device abstraction + fixture replay means Linux/Jetson validation does not re-derive the pipeline.
Newer-model churn (YOLO26, Smart Turn, Qwen3) — young libraries.	API changes or subtle behavioral regressions after upgrades.	Version-pinned requirements; contract tests + fixtures make swaps cheap; golden-file diffs surface behavior changes before they reach the robot.
Scope creep from "bugs worth having as features".	Prototype never lands.	Every CSV item is mapped to exactly one phase (\$12) or explicitly deferred to Phase 7 — the map is the scope control.